

# Joseph-Vu Trinh

571-383-0053 | josephvutrinh.work@gmail.com | linkedin.com/in/josephvutrinh | github.com/josephvutrinh

## Education

### Virginia Tech

Bachelor of Science - Computer Science

Blacksburg, VA

Graduating May 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Systems

## Experience

### Software Developer

June 2026 – Present

Virginia Tech

Blacksburg, VA

- Engineered a native iOS application in **SwiftUI** using the **MVVM** architecture to extend Virginia Tech's DBWorkout SQL learning platform to mobile devices
- Integrated **RESTful APIs**, user authentication, **SQL editing**, progress tracking, **AI-powered hints**, and push notifications to deliver an interactive mobile learning experience
- Collaborated with faculty to define product requirements and deliver a production-ready application supporting approximately 35% of Virginia Tech Computer Science students

### Full Stack Developer Intern

Sep 2024 – Dec 2024

Engauge

College Park, MD

- Revamped the front-end of a classroom web application in **React**, **TypeScript**, and **MUI**, rebuilding component architecture to improve consistency and reduce redundant rendering, used by **150+** students during live sessions
- Shipped a real-time collaborative whiteboard synchronizing state across **30+** concurrent clients using **Firestore** listeners, contributing to a **30%** increase in student engagement during live classroom sessions
- Engineered a **Firestore** data model that scoped whiteboards to individual students while providing instructors real-time visibility across session submissions

### Undergraduate Researcher

Jan 2026 – June 2026

Virginia Tech

Blacksburg, VA

- Engineered components of a machine learning pipeline using **OpenPose** and gaze estimation to classify student engagement from classroom videos, improving classification accuracy by **25%** across **150+** students
- Extracted and tracked pose and gaze features from classroom recordings by processing 180 one-minute video segments
- Validated model outputs against survey-based observations, troubleshooting inconsistencies to improve pipeline reliability

### Web Developer

July 2026 – Present

SASE at Virginia Tech

Blacksburg, VA

- Refactored the organization's **React** codebase by migrating legacy CSS to **Tailwind CSS**, establishing a responsive, mobile-first front-end architecture and improving maintainability
- Implemented **10+** reusable React components while redesigning the FAQ, Contact, and Membership pages to streamline future development
- Collaborated with organization leadership to gather requirements, iterate on designs, and deliver a production-ready web application ahead of deployment

## Projects

**Playlist Curator** | *React, TypeScript, FastAPI, Anthropic API, Spotify Web API, librosa*

- Developed an AI-powered playlist curation platform using **React**, **TypeScript**, **FastAPI**, and the **Spotify Web API** that evaluates tracks against user-defined themes before presenting a GitHub-style diff for users to review and apply changes to live playlists
- Integrated **10+ backend API endpoints** and local audio signal analysis with **librosa** to analyze playlists of arbitrary size using both acoustic features and Spotify metadata

**Professor Search Platform** | *VTHacks | Python, React, Databricks Agent Bricks, GraphQL*

- Engineered Python data pipelines with asynchronous Reddit scraping and paginated **GraphQL** ingestion, applying filtering, deduplication, retry logic, and rate-limit handling to collect **16,000+** records across **577 professors** and **3,267 reviews**
- Integrated the curated dataset with **Databricks Agent Bricks** for an LLM-powered recommendation system answering natural-language questions about professors and courses, and developed the supporting React interface

## Technical Skills

**Languages:** Python, Java, JavaScript, TypeScript, C/C++, SQL, Swift

**Frameworks:** React, React Native, Node.js, FastAPI, SwiftUI, Tailwind CSS

**Tools & Technologies:** Git, Linux/Unix, Firebase/Firestore, MongoDB, Docker, CI/CD, Figma, AWS